

**Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination**

**WIRELESS SENSOR NETWORK**

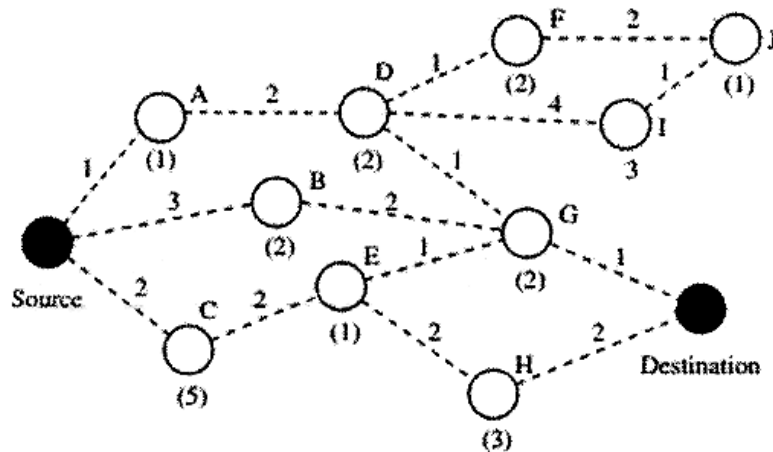
Time : 3 Hours]

[Max. Marks : 60

**Instructions to Candidates :—**

- (1) All questions are compulsory and carry marks as mentioned.
- (2) Draw neat and clean diagram, wherever necessary
- (3) Use of Non-programmable calculator is permitted.
- (4) Assume suitable data wherever necessary.

1. (a) How can oversampling of sensor data overcome the effect of noise ? In structural health monitoring, inspection techniques are classified into global and local. Explain the difference between these techniques. For which of these techniques are wireless sensor network suitable. 5(CO5)
- (b) Compare various routing choices considering different energy metrics for the Figure.



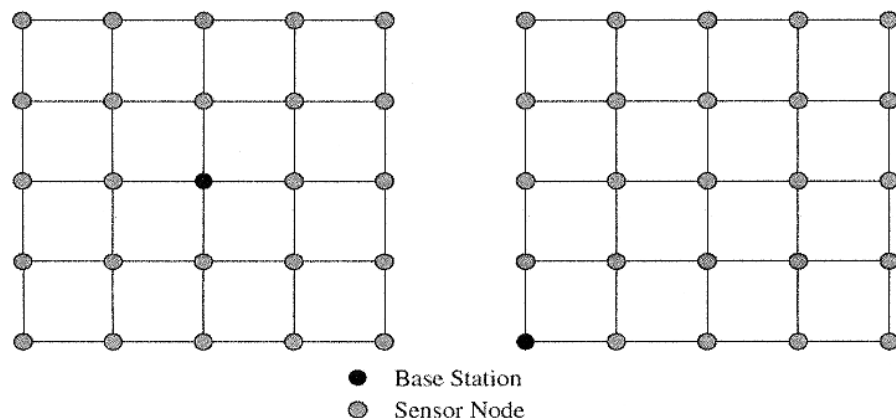
5(CO2)

2. (a) Discuss sensor node components and demonstrate architecture of XYZ sensor node with a neat sketch. 10(CO1)

3. (a) Evaluate the trade-offs between power consumption, performance and design complexity in ultra-low power sensor node design. 5(CO5)
- (b) Discuss the cluster head election policy in the LEACH protocol and explain how LEACH can consider the available energy on each node in this election process. What is the problem with this energy-aware election policy ? Further, LEACH uses TDMA within a cluster, explain the advantages and disadvantages of this approach. 5(CO3)

4. (a) Flooding is a simple strategy for distributing data to one specific node or all sensor nodes in a network. Answer the following questions :

- (a) Explain the three challenges of flooding.
- (b) Which of these can be addressed by gossiping and how can they be addressed ?
- (c) For the topologies shown in Figure, what are good choices for the maximum hop count ? Explain your answer.



- (d) How do sequence numbers contribute to reducing unnecessary transmissions ? Are sequence numbers alone sufficient and if not, what other information is needed to use them correctly ? 10(CO3)

5. (a) In a synchronized network with unknown synchronization error, an anchor node periodically broadcasts an acoustic signal to sensor nodes in its ranges. At time 2000 ms on the anchor node's clock, the anchor node issues a beacon, which is received by node A at time 4000 ms (on node A's clock). What is the distance that A can now compute ? 5(CO4)

- (b) Instead of computing the distance itself, node A also responds with an acoustic signal issued at time 5000 ms, which is received by the anchor node at time 6600 ms. What is the distance computed by the anchor node ? What can you say about the synchronization of anchor node and node A ?  
5(CO4)
6. (a) How does the S-MAC protocol reduce the duty cycles of sensor nodes ? How does the S-MAC protocol attempt to reduce collisions ? How does it address the hidden-terminal problem ? Name at least three disadvantages of the S-MAC protocol.  
5(CO3)
- (b) Compare following routing protocols :
- (i) DSDV,
  - (ii) OLSR,
  - (iii) AODV,
  - (iv) DSR and
  - (v) GPSR.
- 5(CO2)

